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Shell	Principal quantum number (n)	maximum number of electron in a shell (orbital) $2n^2$	Azimuthal quantum number (l) = 0,1.... (n-1)	Maximum no. of electron in a orbital 2 (2l+1)	Magnetic quantum number (m) different possible orientation of orbital	Designation of orbitals in a given shell
M	3	$2(3)^2 = 18$	0	2	0	3s
			1	6	-1, 0, +1	$3p_y$, $3p_z$, $3p_x$,
			2	2[2(2)+1] = 10	-2, -1, 0, +1, +2	$3d_{x^2-y^2}$, $3d_{yz}$, $3d_{z^2}$, $3d_{zx}$, $3d_{xy}$
N	4	$2(4)^2 = 32$	0	2	0	4s
			1	6	-1, 0, +1	$4p_y$, $4p_z$, $4p_x$
			2	10	-2, -1, 0, +1, +2	$4d_{x^2-y^2}$, $4d_{xy}$, $4d_{z^2}$, $4d_{yz}$, $4d_{zx}$
			3	2[2(3)+1] = 14	-3, -2, -1, 0, 1, +1, +2, +3	$f_{y(3x^2-y^2)}$, $f_{z(x^2-y^2)}$, f_{yz^2}, f_{z^3}, f_{xz^2}, f_{xyz}, $f_{x(x^2-3y^2)}$,

The labels on the orbitals, such as p_x, d_{z^2}, f_{xyz} etc. are not associated with specific 'm' values

Evaluate Yourself

3. How many orbitals are possible in the 4th energy level? (n=4)

2.5.1 Shapes of atomic orbitals:

The solution to Schrödinger equation gives the permitted energy values called eigen values and the wave functions corresponding to the eigen values are called atomic orbitals. The solution (Ψ) of the Schrödinger wave equation for one electron system like hydrogen can be represented in the following form in spherical polar coordinates r, θ, φ as,

$$\Psi (r, \theta, \varphi) = R(r).f(\theta).g(\varphi) \text{ ------ } (2.15)$$

(where R(r) is called radial wave function, other two functions are called angular wave functions)

As we know, the Ψ itself has no physical meaning and the square of the wave function $|\Psi|^2$ is related to the probability of finding the electrons within a given volume of space. Let us analyse how $|\Psi|^2$ varies with the distance from nucleus (radial distribution of the probability) and the direction from the nucleus (angular distribution of the probability).

Radial distribution function:

Consider a single electron of hydrogen atom in the ground state for which the quantum numbers are n=1 and